

Worksheet: Pretraining & Fine-tuning (3 classes, one sheet)

CS 5090 · Software Engineering with Generative AI · bring this sheet to all three classes this week; it is collected Friday for completion credit. Discussing with your neighbor is encouraged; write your own answer.

Name: _____

Class 1 · Pretraining

warm-up 1.1 In Project 1 you train a model on raw code, and nobody labeled anything. What is the "label" your model learns from, and where does it come from?

apply 1.2 Take the sentence `Huskies are ready to run`. Write ONE training example (input → target) that each pretraining objective would create from it:

Objective	input → target
Next-token (GPT)	
Masked token (BERT)	

apply 1.3 Scaling laws. Your Project 1 model has about 10M parameters. (a) Using the ≈ 20 tokens-per-parameter rule, how many training tokens should you budget? (b) A classmate says: "I'll train my 10M model on 1B tokens: more is always better." What does the compute-optimal view say, and what would you do with that compute instead?

summarize 1.4 In one or two sentences: why does a model pretrained with *no task labels* help a downstream task that has only 5,000 labeled examples?

Class 2 · Fine-tuning

warm-up 2.1 You have a pretrained GPT-style model and 5,000 labeled bug reports (severe / not severe). List two different ways to get a working classifier, and the main cost of each.

apply 2.2 For each SE task, pick the architecture you would fine-tune (encoder / decoder / encoder–decoder) and say whether a NEW head is needed:

Task	Architecture	New head?
Bug-report severity classification		
Entity tagging over code comments (token level)		
Commit-message generation from diffs		
Code completion		

find the flaw 2.3 Plan A: "tonight I will fully fine-tune a 7B model on my laptop GPU." Plan B: "same laptop, LoRA with rank 8." Which is realistic, and why? Then: your training loss keeps falling but validation loss rises after epoch 2. What happened, and what do you do?

summarize

2.4 Pretraining vs fine-tuning: in one short sentence each, what changes in the **data**, the **objective**, and the **compute**?

Class 3 · The same ideas, in code

predict

3.1 The training loop is about to run its first batch. The vocabulary has 50,257 tokens and the weights are freshly initialized. **Predict the initial loss as a number**, and say where your number comes from.

Hint: what distribution does an untrained softmax produce, and what is the cross-entropy of that distribution against any target?

predict

3.2 The sampling code divides the logits by a temperature T before the softmax. One sentence each: what does $T \rightarrow 0$ do to generation, and what does $T = 2$ do?

find the flaw?

3.3 Loading `AutoModelForSequenceClassification.from_pretrained("bert-base-cased", num_labels=5)` prints: *"Some weights ... were newly initialized. You should probably TRAIN this model..."*. Is something broken? What exactly is newly initialized, and what would happen if you deployed without training?

summarize

3.4 Match each code fragment to the concept it implements (write the letter):

Code fragment	Concept
<code>targets = inputs[:, 1:]</code> (shift by one)	_____
<code>x[mask_positions] = MASK_ID</code> , predict the originals	_____
<code>nn.Linear(hidden, num_labels)</code> added on top	_____
<code>probs = softmax(logits / T)</code>	_____
<code>W_delta = B @ A</code> , base weights frozen	_____

Concepts: (a) task head for fine-tuning (b) LoRA (c) next-token pretraining objective (d) masked-language-model objective (e) temperature decoding

Before you leave: tonight, check question 3.1 against your own Project 1 training log: what was YOUR first recorded loss, and does it match $\ln(\text{vocab size})$?
